TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Taoufik Bakri	Email	taoufik.bakri@tno.nl
Institute Name	Netherlands Organisation for Applied Scientific Research (TNO)			
Address	Anna van Buerenplein 1 NL-2595 DA The Hague The Netherlands			

2. Technology Overview

Title of Proposed Technology	Real-Time, Resilient AI Traffic Decision Support Using 3D Mobility Digital Twins			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Mechanical Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Intelligent Transportation Systems (ITS)	ITS Application Services	Robotics/Automation Machinery	Measurement/Sensor Technologies for Automation
Concept of Proposed Technology	**Proposed Technology Overview** The proposed technology is a real-time, resilient AI Mobility Decision Support System operating on a 3D Digital Twin Earth platform, designed to remain effective under both normal and disrupted traffic conditions. Unlike traditional traffic models, the system combines data-driven AI with physics-based simulations to capture the underlying dynamics of the physical environment and to support robust decision-making in the presence of uncertainty. The platform integrates real-time traffic data with **meteorological and hydrological simulations** to support proactive traffic management, infrastructure resilience, and road safety under both normal and extreme conditions, including unpredicted disruptions.			

Core Challenge: Integrated Data Engineering

The core challenge of the project is the development of a **robust and consistent data engineering architecture** that enables reliable AI decision-making. The system synchronizes heterogeneous data sources into a unified digital representation of the real world:

- **High-Fidelity Inputs:** Real-time data from IoT sensors, traffic systems, and weather forecasts are validated and harmonized to ensure reliable AI performance.
- **Cross-Domain Integration:** Traffic management data is systematically combined with meteorological and flooding models.
- **Physical Consistency:** The digital twin preserves physical constraints to support accurate simulation and prediction.

Technical Innovation: Coupling Physics and AI

The key innovation lies in the **real-time integration** of **Computational Fluid Dynamics (CFD)**, based on the **Navier–Stokes equations**, with **geometry-aware AI models** within a **3D Digital Twin environment**:

- **Real-Time Physics- and Geometry-Informed AI:** The system incorporates road geometry, elevation, slopes, curvature, and surface characteristics into physics-based models that are designed to operate under **real-time constraints**. This allows AI models to interpret how environmental dynamics such as water flow and wind interact with the physical road infrastructure during extreme events.
- **Hybrid Modeling for Real-Time Performance:** To preserve the real-time character of the Digital Twin, the platform combines high-fidelity CFD simulations with **reduced-order models, surrogate models, and AI approximations**, enabling fast yet physically consistent decision support.
- **Adaptive Decision Logic:** Rather than relying on rule-based automation, the AI system applies adaptive reasoning under uncertainty by continuously updating decisions as new data becomes available.
- **Predictive Decision Support:** Multiple operational scenarios—such as signal optimization, dynamic rerouting, and safety interventions—are evaluated in advance, allowing proactive traffic management while maintaining real-time responsiveness.

Why TNO?

TNO bridges the gap between fundamental science and operational reality. We offer a unique ecosystem of independent applied research and high-fidelity mathematical modeling to solve the most pressing challenges in mobility and infrastructure.

- **Bridging Research & Policy**: As a national research organization, we operate at the nexus of academia, industry, and government, delivering scientifically rigorous solutions tailored for real-world policy and operational environments.

- **Deep Technical Expertise**: We specialize in Physics-Informed AI and large-scale Digital Twins, allowing us to create resilient decision-support systems that thrive in complex, data-

	rich environments. • **A Neutral, Reliable Partner**: TNO serves as a trusted orchestrator in the technology transfer process, ensuring that advanced innovations are translated into robust, deployable solutions for traffic resilience, safety, and long-term planning.
Rationale for Joint R&D Project	This project requires a **joint R&D approach** because it integrates multiple domains of expertise, each essential to delivering a fully operational **real-time, data-driven, resilient AI Digital Twin for traffic decision support**: 1. **Multidisciplinary Expertise:** ○ AI and machine learning specialists are needed to design adaptive agents and predictive models. ○ Civil and transportation engineers contribute knowledge of road infrastructure, traffic dynamics, and operational constraints. ○ Meteorology and hydrology experts ensure accurate environmental simulations that affect traffic flow. ○ CFD and physics simulation experts integrate real-world dynamics into the Digital Twin. 2. **Complex System Integration:** Developing a **3D Digital Twin platform** that operates in real-time requires synchronized integration of heterogeneous data sources, simulation engines, and AI reasoning frameworks — a challenge that no single partner can fully achieve alone. 3. **Shared Infrastructure and Resources:** Access to high-fidelity traffic sensors, IoT networks, and computational resources (HPC/edge/cloud) is distributed across partners. A joint project ensures these resources are effectively leveraged without duplication. 4. **Risk and Innovation Management:** By combining complementary strengths, the project can **accelerate innovation** while mitigating technical and operational risks associated with large-scale real-time simulations and physics-informed AI.
Key Words	1. Mobility Optimization 2. AI-Based Decision Support 3. 3D Digital Twin Infrastructure 4. Real-Time Operational Integration 5. Cross-Disciplinary Collaboration

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3) (Proof of Concept)	End of the Project	Level (6) (Pilot-scale Prototype Validation)
R&D Objectives	**Project Objectives:** The objective of this joint R&D project is to develop a real-time, data-driven AI-based Mobility Decision Support System operating on a 3D Digital Twin platform, with a strong focus on physics-informed AI and Computational Fluid Dynamics (CFD). Through this project, the consortium aims to strengthen its technical capabilities in integrating AI, traffic engineering, and physics-based simulations to support safer, more resilient, and more efficient mobility operations under weather-related risks. Specifically, the project pursues the following objectives: 1. **Development of AI-Based Traffic Risk Prediction Agents** To develop AI agents that integrate real-time and forecasted weather, hydrological, and traffic data to predict traffic risks such as flooding, congestion, and safety hazards, while explicitly accounting for physical and geometric road characteristics. 2. **Real-Time 3D Digital Twin Visualization and Simulation** To implement real-time visualization and simulation capabilities within a 3D Digital Twin environment, enabling continuous monitoring of traffic conditions, environmental impacts, and infrastructure status at multiple spatial and temporal scales. 3. **Physics-Informed AI Decision-Making Framework** To develop an AI-based decision-making framework that formalizes how weather-driven and physics-based risks propagate through traffic networks, combining CFD-based insights with data-driven models to translate complex environmental dynamics into actionable operational strategies. 4. **Automated Traffic Control and Detour Strategy Generation** To design and implement automated algorithms for traffic signal control, dynamic rerouting, and detour generation that respond adaptively to predicted risks, ensuring real-time operational effectiveness while maintaining physical and traffic consistency. 5. **Intuitive Decision Support User Interface** To construct an intuitive and scalable user interface that supports both operational decision-makers and policy-level stakeholders, enabling scenario exploration, risk assessment, and informed decision-making through interactive 3D visualizations and explainable AI outputs.			
R&D Target Specifications	**1. Feasible (Core Project Targets)** **a. Traffic risk prediction accuracy under weather impacts:** ≥ 80% classification accuracy for weather-related traffic risk events (e.g. flooding, reduced visibility, congestion onset). **b. Real-time data ingestion and processing latency**: ≤ 5 minutes end-to-end latency from data acquisition to Digital Twin update. **c. Traffic control response time**: ≤ 5 minutes from risk detection to generation of operational control recommendations. **d. Operational scenario evaluation speed**: ≥ 10 alternative control or detour scenarios evaluated per event in near real time. **2. Ambitious (High-Impact but Credible Targets)** **a. Travel time reduction in congestion mitigation simulations:** ≥ 15% average travel time reduction under weather-induced congestion scenarios. **b. Risk anticipation horizon**: ≥ 30–60 minutes advance prediction of weather-driven traffic disruptions using physics-informed models. **c. Accuracy improvement over data-only AI baselines**:			

	≥ 20% improvement in risk prediction performance compared to purely data-driven approaches. d. **3D Digital Twin update frequency**: ≥ 1 update per minute for critical road segments during extreme events. 3. **Moon-Shot Targets:** a. **Sub-minute real-time responsiveness:** ≤ 60 seconds end-to-end latency through hybrid CFD–AI surrogate modeling for selected high-risk zones. b. **Travel time reduction during extreme events:** ≥ 20% reduction in average delay during severe weather scenarios compared to current operational strategies. c. **Autonomous control effectiveness:** ≥ 60% of operational traffic control actions executed autonomously without human intervention during predefined emergency conditions. d. **City-scale Digital Twin scalability**: Real-time operation of the 3D Digital Twin for >90% of a metropolitan road network under normal conditions.
R&D Plan	**Proposed R&D Plan (36 Months):** 1. **Phase 1 (Months 1–6): Data Integration and Conceptual Design of AI Agents** a. **Objectives: Establish the foundation for AI agents and data pipelines.** b. **Key Activities:** i. Define interfaces for weather, traffic, and sensor data, ensuring data quality, reliability, and compatibility with the Digital Twin platform. ii. Design agent-based decision-making models, including rules for risk propagation, adaptive behavior, and scenario evaluation. iii. Develop initial simulation environments to test agent behaviors under synthetic or historical conditions. 2. **Phase 2 (Months 6–18): Development of AI-Based Prediction and Decision Algorithms** a. **Objectives: Build core predictive and decision-making capabilities.** b. **Key Activities:** i. Implement traffic risk forecasting and congestion prediction algorithms, integrating physics-informed AI and CFD insights. ii. Develop automated traffic operation logic, including dynamic signal control and detour generation. iii. Formalize decision logic for AI agents, explicitly handling uncertain and dynamic environmental conditions. iv. Conduct iterative validation using historical traffic and weather datasets to refine agent performance. 3. **Phase 3 (Months 18–30): Integration with 3D Digital Twin Platform and Visualization** a. **Objectives: Enable real-time operation, visualization, and scenario evaluation.** b. **Key Activities:** i. Integrate AI agents and algorithms with the 3D Digital Twin environment. ii. Implement real-time monitoring, simulation, and predictive visualization. iii. Develop interactive visual analytics dashboards for operational and policy decision support. iv. Test system responsiveness under real-time constraints and heterogeneous data inputs. 4. **Phase 4 (Months 30–36): Pilot Demonstration and Performance Validation** a. **Objectives: Demonstrate the system in real operational contexts and quantify impact.** b. **Key Activities:** i. Conduct pilot demonstrations using real traffic and weather data from the Netherlands (or other partner regions). ii. Perform quantitative evaluation of traffic risk prediction, travel time reduction, and operational control effectiveness.

	iii. Gather user feedback from operators and policy stakeholders on the usability and interpretability of the interface. iv. Refine models, algorithms, and visualization tools based on pilot results.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO will play a central and multi-level role in the proposed project, leveraging its expertise across data engineering, AI, CFD, traffic engineering, and policy decision support. Specifically, TNO's contributions will include: • Data Engineering & Fusion: Design and validate pipelines for heterogeneous traffic, weather, and sensor data. • AI & Predictive Modeling: Develop physics-informed AI agents for traffic risk prediction and congestion management. • Computational Fluid Dynamics (CFD): Integrate CFD simulations to capture physical impacts on mobility infrastructure. • Traffic Control & Decision Support: Jointly create automated traffic control algorithms and support operational/policy decisions.
Expected R&R of Korean SME Partner	For the Korean partner, we want to emphasize their strengths in 3D Digital Twin, VR, map-matching, and operational reliability. • 3D Digital Twin & Virtual Reality: Develop and maintain the immersive 3D Earth Digital Twin platform for visualization and simulation. • Map-Matching & Data Alignment: Integrate traffic, weather, and sensor data into the Digital Twin, ensuring accurate geospatial alignment. • Real-Time Operational Management: Enable continuous monitoring, system uptime, and reliable real-time data processing. • Scenario Simulation & Visualization: Support interactive simulation of traffic and environmental scenarios for decision-making. • Interface & Deployment Support: Provide tools and expertise to operationalize the Digital Twin for policy and mobility stakeholders with support from TNO.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes:** **1. Validated Data Pipelines** **a.** High-quality, harmonized traffic, weather, and sensor data streams for AI and Digital Twin use. **2. AI-Based Traffic Risk Prediction Models** **a.** Physics-informed agents capable of predicting congestion, flooding, and safety hazards under real-world conditions. **3. Automated Traffic Control Algorithms** a. Dynamic signal control, detour generation, and adaptive operational strategies. **4. Integrated 3D Digital Twin Platform** a. Real-time operational environment combining traffic, weather, and CFD simulations. b. Interactive visualization and scenario simulation for both operational and policy stakeholders. **5. Decision-Making Framework** a. Formalized AI agent logic linking environmental risks to traffic network operations. b. Tools to translate complex simulations into actionable operational strategies. **6. Pilot Demonstration and Validation** a. Performance evaluation with real traffic data (e.g., Netherlands). **b.** Quantitative KPIs: traffic risk prediction accuracy, latency, travel time reduction, operational effectiveness. **7. User Interface for Decision Support** a. Intuitive dashboard for operators and policymakers, supporting scenario analysis and real-time control. **8. Project Reports and Documentation** a. Comprehensive documentation of methodologies, models, integration, and pilot results.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization